

Sa'ad Farooq

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EDUCATION

University of Waterloo

Bachelor of Mathematics, Combinatorics and Optimization

Relevant Coursework: OOP, Elementary Algorithm Design, Algorithmic Problem-Solving, Logic and Computation, Optimization, Statistics.

Waterloo, Ontario

Sept. 2023 – Present

EXPERIENCE

Software Engineer CxE

Böhringer Ingelheim

April 2026 – Present

Burlington, Ontario

- Built a Snowflake Cortex AI agent by authoring semantic-view YAML metadata across 60+ views and developing a Snowflake-embedded MCP server integrated with PydanticAI, enabling contextual SQL generation, Veeva Vault data retrieval, tool selection, and auditable query responses.
- Developed a Python web-scraping and reporting application that automates a full-day manual data-collection workflow and generates analysis-ready reports, while establishing the project environment with Bitbucket version control, Confluence documentation, and Jira-based task tracking.
- Implemented Jenkins CI/CD pipelines to validate and deploy Snowflake SQL and semantic-model changes across environments, enabling repeatable releases and reducing manual deployment risk.

Full-Stack AR Developer

Learnscape Consulting Inc.

May 2026 – Jul 2026

Markham, Ontario (Remote)

- Engineered a full-stack virtual fashion try-on platform using Next.js, TypeScript, and Supabase, implementing REST APIs, pose-detected AR webcam overlays, and SVG rendering logic for realistic fabric draping and live jewelry try-on.
- Collaborated on a client-facing MVP in an Agile/Scrum team, incorporating product-owner feedback throughout iterative sprints and managing deliverables through a project-tracking platform to meet defined milestones.

CS115 Instructional Support Assistant

University of Waterloo

Sept 2025 – Dec 2025

Waterloo, Ontario

- Built Python static-analysis tools to detect recursion and prohibited API usage while distinguishing helper functions from test code across lambdas, local scopes, and incomplete submissions.
- Engineered an automated grading pipeline in Python and Bash, integrating code analysis, test-result parsing, and MarkUs CLI automation to process hundreds of submissions reproducibly.

Systems & Data Analyst

Linamar Corporation

May 2024 – Aug 2024

Guelph, Ontario

- Automated reporting workflows using Excel VBA and SQL, saving 35+ hours/month and accelerating month-end close timelines by 30%.
- Built Power BI dashboards tracking \$10M+ in accounts payable and operational KPIs, improving reporting accuracy and decision-making speed.
- Processed and reconciled 1,000+ vendor invoices and developed 12-week rolling cash flow models, reducing discrepancies by 22% and optimizing liquidity planning.

PROJECT

GRT Stop Request App | *HTML/CSS, JavaScript, Firebase Realtime Database, GTFS*

Jun 2026 – Present

- Built a mobile-first transit accessibility prototype with passenger and driver interfaces, synchronizing anonymous stop requests, acknowledgments, and request statuses in real time through Firebase.
- Integrated GRT GTFS route and stop data to support route selection, nearby-stop discovery, searchable stops, and deployed the application through Firebase Hosting.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C/C++, Bash/Shell, Groovy, HTML/CSS

Frameworks & Libraries: Angular, Ionic, Storybook, Pandas, NumPy, Scikit-learn, XGBoost, Playwright, Gurobi

Cloud, Data & DevOps: AWS S3, Docker, Jenkins, CI/CD, DataHub, EDP, Power BI, Snowflake

Developer & Collaboration: Git, GitHub, Bitbucket, Jira, Confluence, Linux, Agile/Scrum